

LAB 02: RESERVOIR STAGE NUMERICAL ANALYSIS

Four-Parameter Logistic Curve Fitting & Numerical Integration Report

Model Goodness-of-Fit Summary

R ² (Coefficient of Determination):	0.999141	SSE (Sum of Squared Errors):	0.212572 m ²
s (Standard Error of Estimate):	0.048068 m	Degrees of Freedom (dof):	92
Integrated Area (quad):	67.1744 m·h	Integration Absolute Error:	1.83e-10 m·h

Fitted Parameter Estimates & Statistical Significance

Parameter Description	Estimate	Std Error (SE)	t-Statistic	p-Value
c (Baseline Stage Level)	0.4848	0.0145	33.3380	< 0.0001
a (Asymptotic Capacity Rise)	4.0522	0.0198	205.0676	< 0.0001
k (Logistic Growth Rate)	0.4731	0.0065	72.4385	< 0.0001
t _{inf} (Inflection Point Time)	9.9988	0.0316	316.1196	< 0.0001

Diagnostic Reading & Interpretation of Residuals

The four-parameter logistic model demonstrates exceptional fidelity to the observed reservoir stage measurements, yielding an **R² of 0.999141** and a low standard error of estimate (**s = 0.048068 m**). All four fitted parameters demonstrate high statistical significance with **p-values < 0.0001**.

Residual Behavior Analysis:

- Distribution & Sphericity:** The residuals exhibit small magnitudes bounded within ±0.035 m, showing a balanced mean centered around zero. This indicates unbiased performance without global systematic offset.
- Autocorrelation Patterns:** Minute, structured oscillations occur around the inflection zone (t = 10.00 h). This subtle pattern reflects physical transition phases during peak inflow acceleration before transitioning into asymptotic reservoir stabilization.
- Engineering Conclusion:** The integrated area under the curve equals **67.1744 m·h**. This value represents the cumulative pressure-head duration (meter-hours) acting upon the reservoir boundaries over the 24-hour observation horizon.